

5.5 Surface Integrals Worksheet

1. Evaluate $\iint_{\sigma} xy \, dS$, where σ is the portion of the plane $x + y + z = 1$ in the first octant.

2. Evaluate $\iint_{\sigma} \frac{1}{x^2 + y^2 + 1} dS$, where σ is the portion of the paraboloid $x^2 + y^2 = 2z$ inside the cylinder $x^2 + y^2 = 8$.

3. Evaluate $\iint_{\sigma} z^2 dS$ over the sphere $x^2 + y^2 + z^2 = 4$.

4. Consider the object which has the shape of the spiral ramp communicated as a parametric surface by $\mathbf{r}(u, v) = \langle v \cos(u), v \sin(u), u \rangle$, where $0 \leq u \leq 2\pi$ and $0 \leq v \leq 1$. Suppose that the density of the surface at a point is given by the function $f(x, y, z) = \frac{1}{1 + x^2 + y^2}$. Determine the mass of the object.

5. Let σ be the surface of the solid created by the cone $z = \sqrt{x^2 + y^2}$ below the plane $z = 9$.
- a) Use a surface integral to compute the surface area of σ *without* converting the surface to a parametric form.

b) Parameterize the surface σ .

c) Use a surface integral to compute the surface area σ using the parameterization you found in part (b).

6. Determine the surface area for the portion of the paraboloid $z = 4 - x^2 - y^2$ that sits above the xy -plane.

7. Consider the cylindrical surface created by $x^2 + y^2 = 16$ between the planes $z = 1$ & $z = e$. Suppose that the density of the surface at a point is given by the function $f(x, y, z) = \ln(z)z + \frac{1}{16}(x^2 + y^2) - 1$. Determine the mass of the object.